

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

Mathematics — SSC-I (9th Grade)

Syllabus: Chapter 4 — Factorization and Algebraic Manipulation

Total Marks: 65

Time Allowed: 2h 30m

Version: 1

General Instructions:

1. Write your name, roll number, and section clearly on the answer sheet.
2. **Section A** (MCQs): Attempt **ALL 12** questions. Each carries **1 mark**. Time: 15 minutes.
3. **Section B** (Short Questions): Attempt **ALL 11** pairs, choosing the main question **OR** its alternative. Each carries **3 marks**.
4. **Section C** (Long Questions): Attempt **ALL 4** pairs, choosing the main question **OR** its alternative. Each carries **5 marks**.
5. Use black or blue ink only. Pencil is not allowed except for diagrams.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

Encircle the correct option. All questions are compulsory.

#	Question	A	B	C	D	Answer
1	The factorization of $ka + kb + kc$ is:	$k(a - b + c)$	$k(a + b + c)$	$k(a + b - c)$	$(k + a)(b + c)$	○○○○
2	The factorized form of $a^2 - b^2$ is:	$(a + b)^2$	$(a - b)^2$	$(a + b)(a - b)$	$(a - b)(a - b)$	○○○○
3	The factorized form of $a^4 + a^2b^2 + b^4$ is:	$(a^2 + b^2)^2$	$(a^2 + ab + b^2)(a^2 - ab + b^2)$	$(a^2 - b^2)^2$	$(a + b)^4$	○○○○
4	The factors of $x^2 - 9x + 8$ are:	$(x - 1)(x - 8)$	$(x + 1)(x - 8)$	$(x - 1)(x + 8)$	$(x - 2)(x - 4)$	○○○○
5	$a^3 + b^3$ is equal to:	$(a + b)(a^2 + ab + b^2)$	$(a - b)(a^2 + ab + b^2)$	$(a + b)(a^2 - ab + b^2)$	$(a + b)^3$	○○○○
6	The expression $a^3 + 3a^2b + 3ab^2 + b^3$ is equal to:	$(a - b)^3$	$(a + b)^3$	$(a + b)(a^2 - ab + b^2)$	$(a + b)^2$	○○○○
7	HCF uses the <i>lowest</i> power of each common factor, while LCM uses the:	Lowest power of every factor	Highest power of common factors only	Highest power of every factor	Product of all uncommon factors	○○○○
8	If HCF of two polynomials P and Q is H, then LCM equals:	$P \times Q \times H$	$P \times Q \div H$	$H \div (P \times Q)$	$P + Q - H$	○○○○
9	An algebraic fraction is an expression of the form P/Q where P and Q are polynomials and:	$Q = 0$	$P = 0$	$Q \neq 0$	$P = Q$	○○○○
10	The square root of $4a^2 + 12ab + 9b^2$ is:	$\pm(2a + 3b)$	$\pm(2a - 3b)$	$\pm(4a + 9b)$	$\pm(a + b)$	○○○○
11	To add algebraic fractions with different denominators, we first find the:	HCF of denominators	LCM of denominators	HCF of numerators	LCM of numerators	○○○○
12	Factorization is essentially the <i>opposite</i> of which process?	Division	Subtraction	Multiplication	Integration	○○○○

SECTION B — Short Questions (11 × 3 = 33 Marks)

Attempt ALL 11 questions. Choose either the main question OR its alternative.

Q.	Question	Marks	OR	OR Question	Marks
2(i)	Factorize: $2x^2y^3 - 6x^2y^2 + 2xy^3$	3	OR	Factorize by grouping: $3nx - 3x - 3ny + 3y$	3
2(ii)	Factorize: $x^2 - 6ax + 9a^2 - 16b^2$	3	OR	Factorize: $1 - x^2 - 2xy - y^2$	3
2(iii)	Factorize: $x^4 + 4m^4$ (Type-I form)	3	OR	Factorize: $m^4 + m^2 + 1$	3
2(iv)	Factorize: $y^2 - 7y + 12$	3	OR	Factorize: $3x^2 + 22x - 16$	3
2(v)	Factorize: $x^3 + 27y^3$ (sum of cubes)	3	OR	Factorize: $8a^3 - 125b^3 - 2a + 5b$	3
2(vi)	Factorize: $27a^3 + 1 + 27a^2 + 9a$ (Type-IV form)	3	OR	Factorize: $4x^2(2x - 15) - 25(5 - 6x)$	3
2(vii)	Find the HCF of: $(x + y)^2$ and $x^2 - y^2$	3	OR	Find the HCF of: $(a - b)^3$ and $a^2 - 2ab + b^2$	3
2(viii)	Find the HCF of: $ax^2 + 7ax + 12a$, $ax^2 - 5ax - 24a$ and $2ax^2 + 5ax - 3a$	3	OR	Find the HCF of $x^2 - 49$ and $x^2 - 4x - 21$ by factorization	3
2(ix)	State the relation $LCM \times HCF = P \times Q$. If HCF of two polynomials is $x - 7$ and their LCM is $x^3 - 10x^2 + 11x + 70$, find the product of the two polynomials.	3	OR	Find the LCM of: $x^2 + x - 6$, $x^2 + x - 2$ and $x^2 - 4x + 3$	3
2(x)	Simplify: $(x^2 - 1) \sqrt{x^2 + 4x + 4} \times (x + 2) \sqrt{x^2 + 2x - 3}$	3	OR	Simplify: $(r^2 - s^2) \sqrt{r} \div (r - s) \sqrt{s}$	3
2(xi)	Simplify: $\frac{1}{x(x - y)} + \frac{1}{y(x + y)}$	3	OR	Find the square root of: $4a^2 + 12ab + 9b^2$ by the factorization method.	3

SECTION C — Long Questions (4 × 5 = 20 Marks)

Attempt ALL 4 questions. Choose either the main question OR its alternative.

Q.	Question	Marks	OR	OR Question	Marks
3(i)	Factorize completely: $(x^2 + 3x - 4)(x^2 + 3x + 5) + 8$ Show all substitution steps clearly.	5	OR	Factorize completely: $(x + 5)(x + 3)(x + 2)(x + 6) - 88$ Pair the binomials strategically and use a substitution to simplify.	5
3(ii)	Find the HCF of the following three polynomials using the factorization method: $P = 2m^2 - 2mn$, $Q = 4m^4 - 4m^2n^2$, $R = 2m^3 - 4m^2n + 2mn^2$	5	OR	Find the HCF of the following two polynomials using the division method : $P = x^3 + x^2 - 5x + 3$ and $Q = x^2 + 3x$	5
3(iii)	Find the LCM of the following polynomials: $2x^2 - 12xy + 16y^2$, $x^2 - 6xy + 8y^2$ and $3x^2 - 12y^2$ Hence verify: $LCM \times HCF = \text{Product of any two of the polynomials (using the first two)}$.	5	OR	Product of two polynomials is $x^4 + 3x^3 - 12x^2 - 20x + 48$ and their HCF is $x - 2$. Find their LCM. Also verify $LCM \times HCF = \text{Product of the two polynomials}$.	5
3(iv)	Simplify the following expression completely: $-2x \sqrt{x + 3} + \frac{3}{(3 - x)} - \frac{(8x - 12) \sqrt{x^2 - 9}}{(x^2 - 9)}$	5	OR	Simplify the following expression completely: $(3u^2 + 5uv + 2v^2) \sqrt{u^2 + 5uv + 6v^2} + (2u + 6v) \sqrt{u^2 - 4v^2} \div (u^2 + 6uv + 9v^2) \sqrt{u^2v - 2uv^2}$	5

Invigilator's Signature: _____

Candidate's Signature: _____

Chapters Covered: Ch 4 — Factorization and Algebraic Manipulation

— End of Question Paper —